Detailed Methodology for Section 18: Annual Climate and Year-Round Operating Layer

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1 Detailed Methodology for Section 18: Annual Climate and Year-Round Operating Layer

This section documents, in report form, how I implemented the annual climate and daily duty simulation described by Section 18 (229–251). I wrote this layer to convert monthly climate statistics into a physically consistent daily operating profile, then compute daily heating or cooling duty, bed temperature response, electricity use, cost, and water replacement.

Important modeling clarification (Gaussian terminology). Section 18 uses periodic Gaussian-kernel regression (weighted averaging), not Gaussian Process Regression (GPR). In this implementation, no posterior covariance model is solved and no Bayesian kernel matrix inversion is performed. Instead, each day is computed as a distance-weighted average of monthly reference values using a Gaussian weighting function, with circular (wraparound) day distance.

Notation identity clarifications used in Section 18

To avoid symbol ambiguity, I use the following identities throughout this section:

- m: month index ($m = 1, \ldots, 12$).
- d: day-of-year evaluation index ($d = 1, \ldots, P$).
- P: annual period length in days ($P = 365$ in this implementation).
- d_i: day-of-year reference day associated with reference point i (month midpoint in this implementation).
- d_m: month-midpoint day-of-year for month m. With one reference point per month, d_m and d_i refer to the same reference-day map.
- L_m: number of calendar days in month m.
- i in Eqs. (229)–(230): reference-point index over the monthly climate reference values.
- i in Eqs. (232), (236): day-of-month candidate index inside one month for event placement. This is a different role than i in Eqs. (229)–(230), despite sharing the same symbol.
- y_i: reference value at reference point i. In Section 18, this is used in two separate passes:

- pass A: $y_i = \bar{T}_i$ (monthly mean temperature),
- pass B: $y_i = \sigma_i$ (monthly standard deviation).

- $\hat{y}(d)$: kernel-regressed daily estimate generated from one selected y_i pass at day d.
- $w_i(d)$: Gaussian kernel weight applied to reference point i when evaluating day d.
- h: Gaussian kernel bandwidth in days.
- integer allocation rule: convert fractional monthly expectations into integer monthly counts (freeze and heat-wave both use month-wise stochastic rounding in this walkthrough).
- $\beta_{f,m}$, $\rho_{f,m}$, $U^{\text{cnt}}_{f,m}$: freeze-count floor, freeze-count fractional remainder, and freeze-count uniform draw for month m.
- $\beta_{s,m}$, $\rho_{s,m}$, $U^{\text{cnt}}_{s,m}$: spell-start count floor, fractional remainder, and uniform draw for month m.
- $\beta_{h,m}$, $\rho_{h,m}$, $U^{\text{cnt}}_{h,m}$: heat-wave-day count floor, fractional remainder, and uniform draw for month m.
- N_f, N_s, N_h: realized annual totals after month-wise stochastic rounding (freeze days, spell starts, and heat-wave days).
- $s_{f,m}$, $r_{f,m}$: freeze-day Gaussian spread and repulsion length scales for month m.
- $\mathcal{S}_{f,m}$: set of freeze-day indices already selected in month m.
- $\mathcal{S}^{(k)}_{f,m}$: freeze-day selected set after k recursive selections ($\mathcal{S}^{(0)}_{f,m} = \varnothing$).
- $\mathcal{I}^{(k)}_{f,m}$: freeze-day candidates remaining at step k, defined by $\mathcal{I}^{(k)}_{f,m} = \{1, \ldots, L_m\} \setminus \mathcal{S}^{(k-1)}_{f,m}$.
- $\mathcal{I}$ is a calligraphic capital I (candidate-index set), not L.
- r: generic day-of-month index, $r \in \{1, \ldots, L_m\}$.
- $d_{f,m,k}$: freeze day selected at recursion step k.
- $R^{(k)}_{f,m}(r)$: deterministic ranking value for freeze-day candidate day r at step k; larger R means higher selection priority.
- $M_{f,m}$: freeze-anomaly monthly center value for month m (mean freeze anomaly below $\bar{T}_m$).
- $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$: lower/upper freeze-anomaly bounds for month m.
- $\sigma_{\Delta T,f,m}$: freeze-anomaly spread parameter for month m.
- $\Delta T_{f,m,q}$: sampled anomaly for selected freeze event q in month m (bounded to $[\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}]$).
- $A_{f,m}(r)$: day-of-month freeze-anomaly field; $A_{f,m}(r) = \Delta T_{f,m,q}$ when day r is the q-th selected freeze day, and $A_{f,m}(r) = 0$ otherwise.
- $s_{h,m}$, $r_{h,m}$: heat-wave start Gaussian spread and repulsion length scales for month m.
- $\mathcal{S}_{h,m}$: set of heat-wave start-day indices selected in month m.

- $\mathcal{S}_{h,m}^{(k)}$: heat-wave-start selected set after k recursive selections ($\mathcal{S}_{h,m}^{(0)} = \varnothing$).
- $\mathcal{I}_{h,m}^{(k)}$: heat-wave-start candidates remaining at step k, defined by $\mathcal{I}_{h,m}^{(k)} = \{1, \ldots, L_m\} \setminus \mathcal{S}_{h,m}^{(k-1)}$.
- $d_{h,m,k}$: heat-wave start day selected at recursion step k.
- $R_{h,m}^{(k)}(r)$: deterministic ranking value for heat-wave-start candidate day r at step k; larger R means higher selection priority.
- q: heat-wave spell index within month m, ordered by projected start day.
- $r_m(d)$: day-of-month index corresponding to day-of-year d in month m.
- $\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}$: lower/upper spell-anomaly bounds (above $\bar{T}_m$) for month m, loaded from NOAA-derived monthly spell statistics.
- $\sigma_{\Delta T,h,m}$: configured or fallback standard deviation used for spell-anomaly sampling in month m.
- $\Delta T_{h,m,q}$: sampled anomaly assigned to spell q in month m, bounded to $[\Delta T_{h,m}^{\text{low}}, \Delta T_{h,m}^{\text{high}}]$.
- $M_{h,m,q}$: spell-magnitude variable for spell q in month m; in code-equivalent notation $M_{h,m,q} \equiv \Delta T_{h,m,q}$.
- $X_{h,m,q}$: stochastic spell-anomaly draw (in °C) used when $\kappa \notin \mathcal{D}_{\text{det}}$. It is defined explicitly in Step 18.3-F as the first accepted transformed normal draw in $[a_{h,m}, b_{h,m}]$, or the clipped-center fallback if no acceptance occurs.
- $A_{h,m}(r)$: day-of-month spell-anomaly field for month m, obtained by allocating each spell magnitude $M_{h,m,q}$ to all days in spell block $\mathcal{B}_{h,m,q}$.
- $g_{\text{std}}(z)$, $G_{\text{std}}(z)$: standard-normal density and CDF used in the explicit truncated-normal equations.
- $\varnothing$: empty set; $A \setminus B$: set difference; $A \cup \{x\}$: add element x to set A; $|A|$: set size.
- $D_h(d)$, $D_c(d)$: mode-specific clipped duty fractions.
- $Q_{\text{to bed},h}(T_\infty(d))$: signed net heat-to-bed returned by the heating reference-point re-solve at day ambient $T_\infty(d)$ (positive when the heater adds heat to the bed).
- $Q_{\text{to bed},c}(T_\infty(d))$: signed net heat-to-bed returned by the cooling reference-point re-solve at day ambient $T_\infty(d)$ (typically negative when cooling removes heat from the bed).
- $Q_{h,\text{cap}}(d)$, $Q_{c,\text{cap}}(d)$: positive heating and cooling capacities used in duty equations, with $Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d))$.
- $P_h(T_\infty(d))$: heating-mode electrical power from the day-ambient heating re-solve.
- $P_{\text{spot cooler}}(d)$: spot-cooler electric power on day d from the cooling re-solve.
- $P_{\text{assist blower}}(d)$: assist-blower electric power on day d from the cooling re-solve.
- $P_{h,\text{elec}}(d)$, $P_{c,\text{elec}}(d)$: mode electric powers used in daily energy equations, where $P_{h,\text{elec}}(d) = P_h(T_\infty(d))$ and $P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d)$.

- $D(d)$ in Eq. (251): active duty fraction for the current mode/day (equal to $D_h(d)$ on heating days or $D_c(d)$ on cooling days).
- f_{day} in Eq. (244): generic clipped-duty expression; in code this is implemented as `dutyApplied`.
- $\dot{Q}_{\text{load}}$, $\dot{Q}_{\text{nom}}$ in Eq. (244): generic load and nominal-capacity placeholders corresponding to the active mode (heating or cooling).

Temperature-variable identity and derivation map

The following temperature symbols are all explicit in this implementation:

- $\bar{T}_m$: monthly Connecticut mean ambient from `climate_data.monthlyMean_C`.
- σ_m: monthly ambient spread from `climate_data.monthlyStd_C`.
- $T_{\infty,\text{base}}(d)$: regressed daily ambient baseline from periodic Gaussian kernel regression on $\bar{T}_m$.
- $\sigma(d)$: regressed daily spread from the same kernel pass on σ_m.
- $T_{\infty,\text{preH}}(d)$: intermediate daily ambient after freeze-day cold overrides (input to 18.3 heat-wave overlay).
- $T_{\infty,\text{heatRaw}}(d)$: optional diagnostic track exported by code; mathematically, Eq. 18.3-G is written directly in combined form for $T_\infty(d)$.
- $T_m(d)$: month-local daily ambient profile used during event overlays inside month m.
- $T_\infty(d)$: final day ambient passed to thermal load and capacity re-solves.
- $T_{h,\text{set}}, T_{c,\text{set}}$: resolved heating/cooling setpoints from the annual config (or inherited model defaults).
- $T_b(d), T_t(d)$: bottom/top bed-node temperatures from the two-node steady solve under $Q_{\text{bed}}(d)$ and $T_\infty(d)$.
- For each heat-wave spell q: threshold temperature is $\bar{T}_m + \Delta T_{h,m,q}$, where $\Delta T_{h,m,q}$ is a bounded normal draw.
- NOAA spell-bound files used by config generation:
  - raw daily station rows: `Heat Transfer Study/data/climate/noaa_daily_summaries_USW00014740_2016_`
  - derived monthly freeze bounds: `Heat Transfer Study/data/climate/noaa_freeze_day_anomaly_bounds_`
  - derived monthly spell bounds: `Heat Transfer Study/data/climate/noaa_heatwave_spell_anomaly_bour`
  - extraction utility: `Heat Transfer Study/tools/derive_noaa_heatwave_spell_bounds.py`.

Direct NOAA-derived inputs vs model-computed variables (explicit provenance)

To make data provenance explicit, the Section 18 symbols split into two groups.

A. Direct NOAA-derived reference inputs (loaded from config/file). These are treated as fixed monthly inputs by the year-round model:

- Statewide monthly climate (NOAA CAG monthly series, 2016–2025):

$$\bar{T}_m = \texttt{climate_data.monthlyMean_C}[m], \qquad \sigma_m = \texttt{climate_data.monthlyStd_C}[m].$$

  where $T^{\mathrm{CAG}}_{y,m}$ is the NOAA CAG statewide monthly mean temperature (in °C) for year y, month m, and Y is the number of years in the NOAA window:

$$\bar{T}_m = \frac{1}{Y}\sum_{y=1}^{Y} T^{\mathrm{CAG}}_{y,m},$$

$$\sigma_m = \sqrt{\frac{1}{Y}\sum_{y=1}^{Y}\left(T^{\mathrm{CAG}}_{y,m} - \bar{T}_m\right)^2}.$$

- Station-based freeze statistics (NOAA daily-summaries station USW00014740, 2016–2025):

$$F_m = \texttt{climate_data.freeze.averageDaysPerMonth}[m],$$

  where F_m is the historical mean freeze-day count in month m:

$$F_m = \frac{1}{Y}\sum_{y=1}^{Y} C_{y,m},$$

  with $C_{y,m}$ = number of freeze days observed in year y, month m, and Y = number of years in the NOAA window.

$$\mu_{f,m} = \texttt{climate_data.freeze.meanDayOfMonth}[m],$$

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m],$$

$$\Delta T^{\mathrm{low}}_{f,m} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T^{\mathrm{high}}_{f,m} = \texttt{climate_data.freeze.ma}$$

$$\sigma_{\Delta T,f,m} = \texttt{climate_data.freeze.anomalyStd_C}[m].$$

- Station-based heat-wave timing/count statistics (same station/time window):

$$S_m = \texttt{climate_data.heat_wave.averageSpellStartsPerMonth}[m],$$

$$H_m = \texttt{climate_data.heat_wave.averageDaysPerMonth}[m],$$

$$\mu_{h,m} = \texttt{climate_data.heat_wave.meanStartDayOfMonth}[m].$$

- Station-based heat-wave spell-anomaly bounds (from `noaa_heatwave_spell_anomaly_bounds_USW00014740_2` when `useDerivedSpellAnomalyBounds=true`):

$$\Delta T_{h,m} = \texttt{meanAnomalyAboveMonthlyMean_C}[m],$$

$$\Delta T^{\mathrm{low}}_{h,m} = \texttt{minSpellAnomalyAboveMonthlyMean_C}[m], \quad \Delta T^{\mathrm{high}}_{h,m} = \texttt{maxSpellAnomalyAboveMonthlyMean_C}[m],$$

$$\sigma_{\Delta T,h,m} = \texttt{spellAnomalyStd_C}[m].$$

B. Model-computed variables (not direct NOAA inputs). These are produced by equations in Section 18 from the NOAA-derived inputs above:

- kernel-regressed daily fields $T_{\infty,\text{base}}(d)$, $\sigma(d)$,
- allocated integer counts $n_{f,m}$, $n_{s,m}$, $n_{h,m}$,
- selected index sets/lists $\mathcal{S}_{f,m}$, $\mathcal{S}_{h,m}$, and spell blocks $\mathcal{B}_{h,m,q}$,
- sampled spell anomalies $\Delta T_{h,m,q}$,
- final ambient $T_\infty(d)$, duties $D_h(d), D_c(d)$, loads, energy, cost, and water outputs.

So, NOAA provides monthly/statistical inputs; Section 18 equations transform those into daily operational trajectories.

18.1 Climate reference values and regression baseline (229, 230)

I began with monthly Connecticut reference values:

- monthly mean temperature $\bar{T}_m$,
- monthly standard deviation σ_m,
- month lengths from the annual configuration.

I mapped each month to a day-of-year reference day at the month midpoint and evaluated all 365 representative days. For each day d, I computed a Gaussian-weighted average:

$$\hat{y}(d) = \frac{\sum_i w_i(d) y_i}{\sum_i w_i(d)}, \qquad w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right],$$

with circular distance

$$\delta_i(d) = \min(|d - d_i|,\ P - |d - d_i|)\,,$$

where $P = 365$ days and h is the configured bandwidth (default 18 days). I applied this same regression twice:

1. once to monthly means to get daily baseline ambient $T_{\infty,\text{base}}(d)$,
2. once to monthly standard deviations to get daily $\sigma(d)$ for climate-context reporting.

Explicitly, the baseline ambient used by Sections 18.2–18.5 is

$$T_{\infty,\text{base}}(d) = \frac{\sum_{i=1}^{12} w_i(d)\,\bar{T}_i}{\sum_{i=1}^{12} w_i(d)},$$

with

$$w_i(d) = \exp\left[-\frac{1}{2}\left(\frac{\delta_i(d)}{h}\right)^2\right], \qquad \delta_i(d) = \min(|d - d_i|,\ 365 - |d - d_i|)\,.$$

Here d_i is month-i midpoint day-of-year and $\bar{T}_i$ is NOAA monthly mean for month i. In other words, y_i is not a third intermediary variable family; it is whichever reference-value array is active in the current regression pass.

The circular distance is essential because it treats late December and early January as neighboring points, preventing an artificial year-boundary discontinuity.

18.2 Freeze-day allocation and anomaly override (231–234)

This subsection introduces the explicit set-recursion logic used for event-day placement in both 18.2 and 18.3.

The NOAA-derived freeze expectation in each month is fractional (for example, 26.3 days). I convert it to an integer with month-wise stochastic rounding:

$$\beta_{f,m} = \lfloor F_m \rfloor, \qquad \rho_{f,m} = F_m - \lfloor F_m \rfloor.$$

$$U^{\text{cnt}}_{f,m} \sim \text{Uniform}(0,1).$$

$$n_{f,m} = \begin{cases} \beta_{f,m} + 1, & U^{\text{cnt}}_{f,m} < \rho_{f,m}, \\ \beta_{f,m}, & \text{otherwise.} \end{cases}$$

For this rule,

$$\mathbb{E}[n_{f,m}] = F_m$$

and each month is rounded independently. The annual freeze total

$$N_f = \sum_{m=1}^{12} n_{f,m}$$

is therefore random (close to $\sum_m F_m$ in expectation, but not forced to equal round($\sum_m F_m$)). So month m receives exactly $n_{f,m}$ freeze days for that realization.

Step 18.2-A: Define the candidate universe and initialization.

$$\mathcal{U}_m = \{1, 2, \ldots, L_m\}, \qquad \mathcal{S}^{(0)}_{f,m} = \varnothing.$$

Here:

- $\mathcal{U}_m$ is the full set of day-of-month candidates,
- L_m is the number of days in month m,
- $\varnothing$ is the empty set.

Step 18.2-B: Define the remaining candidates at selection step k.

$$\mathcal{I}^{(k)}_{f,m} = \mathcal{U}_m \setminus \mathcal{S}^{(k-1)}_{f,m}, \qquad k = 1, \ldots, n_{f,m}.$$

$\setminus$ means set difference: remove already-selected days. In the current implementation, the freeze-selection spread and repulsion scales are

$$s_{f,m} = \max\left(2, \frac{L_m}{6}\right), \qquad r_{f,m} = \max\left(1.5, \frac{L_m}{14}\right),$$

with L_m the number of days in month m.

Step 18.2-C: Compute deterministic rank value for each remaining candidate.

$$W_{f,m}(r) = \exp\left[-\frac{(r-\mu_{f,m})^2}{2s_{f,m}^2}\right],$$

$$R_{f,m}^{(k)}(r) = W_{f,m}(r) \prod_{p\in\mathcal{S}_{f,m}^{(k-1)}} \left[1 - 0.65\exp\left(-\frac{(r-p)^2}{2r_{f,m}^2}\right)\right], \qquad r \in \mathcal{I}_{f,m}^{(k)}.$$

The factor $W_{f,m}(r)$ prefers days near mean freeze day $\mu_{f,m}$. The product term repels new selections from previously selected days. So $R_{f,m}^{(k)}(r)$ is a ranking value, not a probability. Its direct effect on the selected set is:

$$M_{f,m}^{(k)} = \max_{r\in\mathcal{I}_{f,m}^{(k)}} R_{f,m}^{(k)}(r),$$

$$d_{f,m,k} \in \mathcal{I}_{f,m}^{(k)}, \qquad R_{f,m}^{(k)}(d_{f,m,k}) = M_{f,m}^{(k)},$$

$$\mathcal{S}_{f,m}^{(k)} = \mathcal{S}_{f,m}^{(k-1)} \cup \{d_{f,m,k}\}.$$

So at step k, $R_{f,m}^{(k)}$ determines exactly which new index enters $\mathcal{S}_{f,m}^{(k)}$.

Step 18.2-D: Select one day and update by union.

$$d_{f,m,k} \in \mathcal{I}_{f,m}^{(k)}, \qquad R_{f,m}^{(k)}(d_{f,m,k}) \geq R_{f,m}^{(k)}(r) \text{ for every } r \in \mathcal{I}_{f,m}^{(k)},$$

$$\mathcal{S}_{f,m}^{(k)} = \mathcal{S}_{f,m}^{(k-1)} \cup \{d_{f,m,k}\}.$$

$\cup$ means set union: add the newly selected day index. Because $d_{f,m,k} \in \mathcal{I}_{f,m}^{(k)}$, it is not already in $\mathcal{S}_{f,m}^{(k-1)}$. Therefore each update adds exactly one new day, so $|\mathcal{S}_{f,m}^{(k)}| = k$.

After $n_{f,m}$ steps:

$$\mathcal{S}_{f,m} = \mathcal{S}_{f,m}^{(n_{f,m})},$$

so $\mathcal{S}_{f,m}$ contains exactly $n_{f,m}$ distinct freeze days.

Step 18.2-E: Same logic without set notation (code-equivalent view). If set notation is unfamiliar, the same selection can be written as an index-list loop:

```
selected = []                                # freeze day indices already chosen
for k in 1..n_f,m:
    best_r = None
    best_score = -infinity
    for r in 1..L_m:
        if r in selected:
            continue
        score = exp(-((r-mu_f,m)^2)/(2*s_f,m^2))
        for p in selected:
            score *= (1 - 0.65*exp(-((r-p)^2)/(2*r_f,m^2)))
        if score > best_score:
            best_score = score
            best_r = r
    selected.append(best_r)
```

This is mathematically identical to the set form:

- "$r \in \mathcal{I}_{f,m}^{(k)}$" means "$r$ not in `selected`",
- "$\mathcal{S}_{f,m}^{(k)} = \mathcal{S}_{f,m}^{(k-1)} \cup \{d_{f,m,k}\}$" means "`selected.append(best_r)`".

Step 18.2-F: Freeze-magnitude definition and day allocation. Define freeze-anomaly monthly center:

$$M_{f,m} \equiv \Delta T_{f,m}.$$

How $M_{f,m}$, bounds, and spread are computed from NOAA data (conceptual derivation):

- Source rows: NOAA daily-summaries station `USW00014740` over 2016–2025.
- Freeze-day definition: $T_{\min} \leq 32°\mathrm{F}$.
- Daily mean proxy in °C: $T_{\mathrm{day}}^{(r)}$ from T_{AVG} if present, otherwise $0.5(T_{\max} + T_{\min})$, converted to °C.
- For each freeze day r in month m, compute cold anomaly below monthly mean:

$$a_{m,r} = \max\Big(\bar{T}_m - T_{\mathrm{day}}^{(r)},\, 0\Big)\,.$$

- Monthly center is the average freeze-day anomaly:

$$M_{f,m} = \frac{1}{N_{f,m}} \sum_{r=1}^{N_{f,m}} a_{m,r},$$

  where $N_{f,m}$ is the number of freeze days used in month m.

- Monthly bounds and spread are:

$$\Delta T_{f,m}^{\mathrm{low}} = \min_r a_{m,r}, \quad \Delta T_{f,m}^{\mathrm{high}} = \max_r a_{m,r}, \quad \sigma_{\Delta T,f,m} = \sqrt{\frac{1}{N_{f,m}} \sum_{r=1}^{N_{f,m}} (a_{m,r} - M_{f,m})^2}.$$

  In this repository, these monthly arrays are precomputed and loaded from:

$$M_{f,m} = \texttt{climate_data.freeze.meanAnomalyBelowMonthlyMean_C}[m].$$

$$\Delta T_{f,m}^{\mathrm{low}} = \texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \quad \Delta T_{f,m}^{\mathrm{high}} = \texttt{climate_data.freeze.maxAr}$$

  Define string-normalization map

$$\mathrm{norm}(s) = \mathrm{lower}\Big(\,\mathrm{replace}\,\big(\,\mathrm{replace}(s, \texttt{"-"}, \texttt{"_"}), \texttt{" "}, \texttt{"_"}\big)\Big).$$

Define freeze model key

$$\kappa_f = \mathrm{norm}(\texttt{freeze.anomalySamplingModel})\,.$$

Define explicit mode flags (no set notation):

$$\texttt{isDeterministic}_f = \begin{cases} 1, & \kappa_f = \texttt{off} \text{ or } \texttt{none} \text{ or } \texttt{disabled} \text{ or } \texttt{deterministic} \text{ or } \texttt{deterministic_mean}, \\ 0, & \text{otherwise}, \end{cases}$$

$$\texttt{isTriangular}_f = \begin{cases} 1, & \kappa_f = \texttt{triangular} \text{ or } \texttt{triangular_mean_matched} \text{ or } \texttt{triangular_between_monthly_min_} \\ 0, & \text{otherwise}. \end{cases}$$

Conceptually:

- κ_f is not a temperature; it is a text switch read from configuration.
- norm$(\cdot)$ just standardizes spelling so equivalent labels map to one key (for example, `Triangular-Mean Matched` and `triangular_mean_matched` become the same key).
- $\texttt{isDeterministic}_f = 1$ means "use one fixed freeze anomaly for all selected freeze days in month m." Then $\Delta T_{f,m,q} = \mu^\star_{f,m}$.
- $\texttt{isTriangular}_f = 1$ means "sample each selected freeze day from the triangular distribution on $[a_{f,m}, b_{f,m}]$." Then $\Delta T_{f,m,q} = X_{f,m,q}$.
- These are logic switches (yes/no), not climate variables.

Define freeze bounds and bounded monthly center:

$$a_{f,m} = \Delta T^{\text{low}}_{f,m}, \quad b_{f,m} = \Delta T^{\text{high}}_{f,m}.$$

These are anomaly-magnitude bounds (in °C), not temperatures. In code/config, they come from

$$\texttt{climate_data.freeze.minAnomalyBelowMonthlyMean_C}[m], \qquad \texttt{climate_data.freeze.maxAnomalyBelowMont}$$

Historical NOAA derivation for month m: for each historical freeze day j, define anomaly

$$a_{m,j} = \max\Big(0, \bar{T}_m - T^{\text{freeze day}}_{m,j}\Big),$$

then

$$\Delta T^{\text{low}}_{f,m} = \min_j a_{m,j}, \qquad \Delta T^{\text{high}}_{f,m} = \max_j a_{m,j}.$$

$$\mu^\star_{f,m} = \begin{cases} a_{f,m}, & M_{f,m} < a_{f,m}, \\ M_{f,m}, & a_{f,m} \le M_{f,m} \le b_{f,m}, \\ b_{f,m}, & M_{f,m} > b_{f,m}. \end{cases}$$

So $\mu^\star_{f,m}$ is the bounded (clipped) freeze-anomaly center for month m.

$$c^{\text{raw}}_{f,m} = 3\mu^\star_{f,m} - a_{f,m} - b_{f,m},$$

where $c^{\text{raw}}_{f,m}$ is the unconstrained mode target for the triangular draw. It is chosen so that the triangular mean $(a_{f,m} + b_{f,m} + c)/3$ equals $\mu^\star_{f,m}$ before bounds enforcement. Why: this keeps stochastic day-level draws centered on the intended monthly anomaly level instead of drifting warmer or colder on average.

$$c_{f,m} = \begin{cases} a_{f,m}, & c^{\text{raw}}_{f,m} < a_{f,m}, \\ c^{\text{raw}}_{f,m}, & a_{f,m} \le c^{\text{raw}}_{f,m} \le b_{f,m}, \\ b_{f,m}, & c^{\text{raw}}_{f,m} > b_{f,m}. \end{cases}$$

Define triangular density on $[a_{f,m}, b_{f,m}]$:

$$t_{f,m}(x) = \begin{cases} \dfrac{2(x - a_{f,m})}{(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & a_{f,m} \le x \le c_{f,m}, \\ \dfrac{2(b_{f,m} - x)}{(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & c_{f,m} \le x \le b_{f,m}, \\ 0, & \text{otherwise.} \end{cases}$$

For each selected freeze event $q = 1, \ldots, n_{f,m}$, allocate event magnitude:

$$\Delta T_{f,m,q} = \begin{cases} \mu^\star_{f,m}, & \texttt{isDeterministic}_f = 1, \\ X_{f,m,q}, & \texttt{isTriangular}_f = 1 \text{ and } b_{f,m} > a_{f,m}, \\ \mu^\star_{f,m}, & b_{f,m} \le a_{f,m}, \\ \mu^\star_{f,m} \text{ or legacy bounded-normal draw}, & \texttt{isDeterministic}_f = 0 \text{ and } \texttt{isTriangular}_f = 0, \end{cases}$$

with explicit triangular generation:

$$U_{f,m,q} \sim \text{Uniform}(0,1), \qquad p_{f,m} = \frac{c_{f,m} - a_{f,m}}{b_{f,m} - a_{f,m}}.$$

$$X_{f,m,q} = \begin{cases} a_{f,m} + \sqrt{U_{f,m,q}(b_{f,m} - a_{f,m})(c_{f,m} - a_{f,m})}, & 0 \le U_{f,m,q} \le p_{f,m}, \\ b_{f,m} - \sqrt{(1 - U_{f,m,q})(b_{f,m} - a_{f,m})(b_{f,m} - c_{f,m})}, & p_{f,m} < U_{f,m,q} \le 1. \end{cases}$$

Then enforce bounds explicitly:

$$\Delta T_{f,m,q} = \begin{cases} a_{f,m}, & \Delta T_{f,m,q} < a_{f,m}, \\ \Delta T_{f,m,q}, & a_{f,m} \le \Delta T_{f,m,q} \le b_{f,m}, \\ b_{f,m}, & \Delta T_{f,m,q} > b_{f,m}. \end{cases}$$

For any $a_{f,m} \le u \le v \le b_{f,m}$,

$$\Pr(u \le X_{f,m,q} \le v) = \int_u^v t_{f,m}(x)\, dx.$$

Define day-of-month freeze-anomaly allocation (piecewise, no indicator notation):

$$\text{Sort selected freeze days as } d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}.$$

$$A_{f,m}(r) = \begin{cases} 0, & r \notin \mathcal{S}_{f,m}, \\ \Delta T_{f,m,q}, & r = d_{f,m,q} \text{ for some } q \in \{1, \ldots, n_{f,m}\}, \end{cases} \qquad r = 1, \ldots, L_m.$$

So selected freeze days can have different magnitudes (stochastic mode) or identical magnitudes (deterministic mode), depending on κ_f. Important: $A_{f,m}(r)$ itself is not a random sampler; it is a mapping from selected day index r to the already-generated value $\Delta T_{f,m,q}$. Any randomness occurs upstream when generating $\Delta T_{f,m,q}$ (when $\texttt{isTriangular}_f = 1$); then q is matched to ordered selected days $d_{f,m,1} < \cdots < d_{f,m,n_{f,m}}$ without extra random reshuffling.

Allocation matters because selection and magnitude are different tasks:

- selection ($\mathcal{S}_{f,m}$) decides *which* days are freeze days,
- allocation ($A_{f,m}(r)$) decides *what anomaly value* each selected day gets.

Step 18.2-G: Apply freeze override on selected days. Let $r_m(d) \in \{1, \ldots, L_m\}$ be the day-of-month corresponding to day-of-year d in month m. Then

$$T^{\text{th}}_{f,m}(d) = \bar{T}_m - A_{f,m}(r_m(d)).$$

Interpretation: $T^{\text{th}}_{f,m}(d)$ is the freeze-threshold ambient for day d in month m. It is the day-specific clamp value generated from the monthly mean $\bar{T}_m$ and allocated freeze anomaly $A_{f,m}(r_m(d))$.

$$T_{\infty,\text{preH}}(d) = \begin{cases} T_{\infty,\text{base}}(d), & r_m(d) \in \mathcal{S}_{f,m} \text{ and } T_{\infty,\text{base}}(d) \le T^{\text{th}}_{f,m}(d), \\ T^{\text{th}}_{f,m}(d), & r_m(d) \in \mathcal{S}_{f,m} \text{ and } T_{\infty,\text{base}}(d) > T^{\text{th}}_{f,m}(d), \\ T_{\infty,\text{base}}(d), & r_m(d) \notin \mathcal{S}_{f,m}. \end{cases}$$

So freeze days are forced colder (or equal) relative to the baseline, never warmer.

Step 18.2-H: Why allocation matters for implementation. The override reads a day-level anomaly field $A_{f,m}(r)$, not just one month scalar. This preserves the selected-day pattern and supports both deterministic and stochastic freeze magnitudes.

18.3 Heat-wave allocation and anomaly override (235–238)

Section 18.3 uses the same set-recursion logic defined in 18.2. The only change is parameter substitution from freeze-day placement to heat-wave start placement:

$$\{n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m}, \mathcal{S}_{f,m}\} \to \{n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m}, \mathcal{S}_{h,m}\}.$$

So the meanings of $\setminus$, $\cup$, and "already selected" are identical to 18.2. If preferred, use the exact same code-style loop from Step 18.2-E and replace $(n_{f,m}, \mu_{f,m}, s_{f,m}, r_{f,m})$ with $(n_{s,m}, \mu_{h,m}, s_{h,m}, r_{h,m})$.

Step 18.3-A: Integer monthly counts. Start from fractional monthly expectations S_m (spell starts) and H_m (heat-wave days). Use the same month-wise stochastic rounding rule as 18.2:

$$\beta_{s,m} = \lfloor S_m \rfloor, \qquad \rho_{s,m} = S_m - \lfloor S_m \rfloor,$$

$$\beta_{h,m} = \lfloor H_m \rfloor, \qquad \rho_{h,m} = H_m - \lfloor H_m \rfloor.$$

$$U^{\text{cnt}}_{s,m} \sim \text{Uniform}(0,1), \qquad U^{\text{cnt}}_{h,m} \sim \text{Uniform}(0,1).$$

$$n_{s,m} = \begin{cases} \beta_{s,m} + 1, & U^{\text{cnt}}_{s,m} < \rho_{s,m}, \\ \beta_{s,m}, & \text{otherwise}, \end{cases}$$

$$n_{h,m} = \begin{cases} \beta_{h,m} + 1, & U^{\text{cnt}}_{h,m} < \rho_{h,m}, \\ \beta_{h,m}, & \text{otherwise}. \end{cases}$$

For this rule,

$$\mathbb{E}[n_{s,m}] = S_m, \qquad \mathbb{E}[n_{h,m}] = H_m.$$

Annual totals

$$N_s = \sum_{m=1}^{12} n_{s,m}, \qquad N_h = \sum_{m=1}^{12} n_{h,m}$$

are random (unbiased in expectation), not fixed by a global remainder redistribution. $n_{s,m}$ is number of spells; $n_{h,m}$ is total heat-wave days.

Step 18.3-B: Spell lengths. Let $q = 1, \ldots, n_{s,m}$ index spells in month m. The code applies feasibility limits first:

$$n_{s,m} \leftarrow \min\left(n_{s,m}, \left\lfloor \frac{L_m + 1}{4} \right\rfloor\right).$$

This bound comes from the enforced structure "spell + at least one non-spell day gap" with minimum spell length 3. Given this $n_{s,m}$, total heat-wave days are clipped to feasible range:

$$n_{h,m}^{\min} = 3n_{s,m}, \qquad n_{h,m}^{\max} = L_m - (n_{s,m} - 1),$$

$$n_{h,m} \leftarrow \min\big(\max(n_{h,m}, n_{h,m}^{\min}),\, n_{h,m}^{\max}\big).$$

Now initialize every spell with the minimum:

$$\ell_{h,m,q} = 3, \qquad q = 1, \ldots, n_{s,m}.$$

Define extra days:

$$e_m = n_{h,m} - 3n_{s,m} \ (\geq 0).$$

The implementation distributes e_m cyclically across spells; this is equivalent to quotient–remainder form:

$$\text{for } n_{s,m} > 0: \qquad u_m = \left\lfloor \frac{e_m}{n_{s,m}} \right\rfloor, \qquad v_m = e_m - u_m n_{s,m},$$

$$\ell_{h,m,q} = \begin{cases} 3 + u_m + 1, & q \leq v_m, \\ 3 + u_m, & q > v_m, \end{cases} \qquad q = 1, \ldots, n_{s,m}.$$

So

$$\boldsymbol{\ell}_{h,m} = (\ell_{h,m,1}, \ldots, \ell_{h,m,n_{s,m}}), \qquad \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

Example: if $n_{s,m} = 3$ and $n_{h,m} = 14$, then $e_m = 5$, $u_m = 1$, $v_m = 2$, and

$$(\ell_{h,m,1}, \ell_{h,m,2}, \ell_{h,m,3}) = (5, 5, 4).$$

Edge case: if $n_{s,m} = 0$, then no spells are formed and the month has no heat-wave spell-length vector.

Step 18.3-C: Start-day set construction (same recursion as 18.2).

$$\mathcal{U}_m = \{1, \ldots, L_m\}, \qquad \mathcal{S}_{h,m}^{(0)} = \varnothing.$$

For $k = 1, \ldots, n_{s,m}$:

$$\mathcal{I}_{h,m}^{(k)} = \mathcal{U}_m \setminus \mathcal{S}_{h,m}^{(k-1)},$$

$$W_{h,m}(r) = \exp\left[-\frac{(r - \mu_{h,m})^2}{2s_{h,m}^2}\right],$$

$$R_{h,m}^{(k)}(r) = W_{h,m}(r) \prod_{p \in \mathcal{S}_{h,m}^{(k-1)}} \left[1 - 0.65 \exp\left(-\frac{(r-p)^2}{2r_{h,m}^2}\right)\right], \qquad r \in \mathcal{I}_{h,m}^{(k)},$$

where $R_{h,m}^{(k)}(r)$ is a deterministic ranking value (not a probability).

$$d_{h,m,k} \in \mathcal{I}_{h,m}^{(k)}, \qquad R_{h,m}^{(k)}(d_{h,m,k}) \geq R_{h,m}^{(k)}(r) \text{ for every } r \in \mathcal{I}_{h,m}^{(k)},$$

$$\mathcal{S}_{h,m}^{(k)} = \mathcal{S}_{h,m}^{(k-1)} \cup \{d_{h,m,k}\}.$$

Final selected start set:

$$\mathcal{S}_{h,m} = \mathcal{S}_{h,m}^{(n_{s,m})}.$$

Step 18.3-D: Enforce non-overlap and month bounds. Raw starts from Step 18.3-C are projected to feasible starts $s_{h,m,q}$. Let $d^{\text{raw}}_{h,m,1} < \cdots < d^{\text{raw}}_{h,m,n_{s,m}}$ be the ordered elements of $\mathcal{S}_{h,m}$. So $d^{\text{raw}}_{h,m,q}$ means: the q-th ordered raw start day before feasibility clipping. Equivalently,

$$d^{\text{raw}}_{h,m,q} = \text{SortAsc}(\mathcal{S}_{h,m})_q, \qquad q = 1, \ldots, n_{s,m}.$$

Code-equivalent path:

$$\mathcal{S}_{h,m} \xrightarrow{\text{sort}} (d^{\text{raw}}_{h,m,1}, \ldots, d^{\text{raw}}_{h,m,n_{s,m}}) \xrightarrow{\text{clip to } [e_{h,m,q}, u_{h,m,q}]} (s_{h,m,1}, \ldots, s_{h,m,n_{s,m}}).$$

Using spell index $q = 1, \ldots, n_{s,m}$:

$$e_{h,m,q} = \begin{cases} 1, & q = 1, \\ s_{h,m,q-1} + \ell_{h,m,q-1} + 1, & q \geq 2, \end{cases}$$

$$u_{h,m,q} = L_m - \left(\sum_{p=q}^{n_{s,m}} \ell_{h,m,p} + (n_{s,m} - q) \right) + 1,$$

$$s_{h,m,q} = \begin{cases} e_{h,m,q}, & d^{\text{raw}}_{h,m,q} < e_{h,m,q}, \\ d^{\text{raw}}_{h,m,q}, & e_{h,m,q} \leq d^{\text{raw}}_{h,m,q} \leq u_{h,m,q}, \\ u_{h,m,q}, & d^{\text{raw}}_{h,m,q} > u_{h,m,q}. \end{cases}$$

Spell q occupies

$$\mathcal{B}_{h,m,q} = \{\, r : \; s_{h,m,q} \leq r \leq s_{h,m,q} + \ell_{h,m,q} - 1 \,\},$$

where r is day-of-month index. So $\mathcal{B}_{h,m,q}$ is the spell-day block (not a start-day set, and not a matrix), with

$$|\mathcal{B}_{h,m,q}| = \ell_{h,m,q}.$$

All heat-wave days in month m are

$$\mathcal{B}^{\text{all}}_{h,m} = \bigcup_{q=1}^{n_{s,m}} \mathcal{B}_{h,m,q}.$$

Because blocks are non-overlapping by construction,

$$\left| \mathcal{B}^{\text{all}}_{h,m} \right| = \sum_{q=1}^{n_{s,m}} |\mathcal{B}_{h,m,q}| = \sum_{q=1}^{n_{s,m}} \ell_{h,m,q} = n_{h,m}.$$

Step 18.3-E: Spell-magnitude definition and distribution-based allocation. Define spell magnitude for each spell q as

$$M_{h,m,q} \equiv \Delta T_{h,m,q}.$$

Also define the distribution-method key κ from `heat_wave.spellAnomalySamplingModel`. In the implementation:

$$\mathcal{D}_{\text{det}} = \{\texttt{off}, \texttt{none}, \texttt{disabled}, \texttt{deterministic}, \texttt{deterministic_mean}\}.$$

Define bounds and bounded parameters:

$$a_{h,m} = \Delta T^{\text{low}}_{h,m}, \quad b_{h,m} = \Delta T^{\text{high}}_{h,m}.$$

Here $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, and $\Delta T^{\text{high}}_{h,m}$ are spell-anomaly-above-monthly-mean statistics (in °C), from

`heat_wave.meanAnomalyAboveMonthlyMean_C`$[m]$, `heat_wave.minSpellAnomalyAboveMonthlyMean_C`$[m]$, `he`

Historical NOAA derivation (month m, spells indexed by q): if $\mathcal{B}_{h,m,q}$ is the set of days in historical spell q (grouped by spell start month m), define

$$a^{(h)}_{m,q} = \frac{1}{|\mathcal{B}_{h,m,q}|} \sum_{t \in \mathcal{B}_{h,m,q}} \max\Big(0,\ T^{\text{avg}}_t - \bar{T}_{m(t)}\Big),$$

where T^{avg}_t is NOAA daily mean proxy (TAVG if present, else $(TMAX + TMIN)/2$) and $m(t)$ is the calendar month index of day t. Thus $\bar{T}_{m(t)}$ is the same monthly-mean notation $\bar{T}_m$, evaluated at day t's month. Then

$$\Delta T_{h,m} = \frac{1}{N_m} \sum_{q=1}^{N_m} a^{(h)}_{m,q}, \qquad \Delta T^{\text{low}}_{h,m} = \min_q a^{(h)}_{m,q}, \qquad \Delta T^{\text{high}}_{h,m} = \max_q a^{(h)}_{m,q}.$$

$$\mu^\star_{h,m} = \begin{cases} a_{h,m}, & \Delta T_{h,m} < a_{h,m}, \\ \Delta T_{h,m}, & a_{h,m} \le \Delta T_{h,m} \le b_{h,m}, \\ b_{h,m}, & \Delta T_{h,m} > b_{h,m}, \end{cases}$$

$$\sigma^\star_{h,m} = \begin{cases} \sigma_{\Delta T,h,m}, & \sigma_{\Delta T,h,m} \ge \varepsilon, \\ \varepsilon, & \sigma_{\Delta T,h,m} < \varepsilon. \end{cases}$$

$\sigma_{\Delta T,h,m}$ is the month-m spell-anomaly spread from `heat_wave.spellAnomalyStd_C`$[m]$. If that value is missing or nonpositive, the implementation fallback is

$$\sigma_{\Delta T,h,m} = \begin{cases} (\Delta T^{\text{high}}_{h,m} - \Delta T^{\text{low}}_{h,m})/4, & \Delta T^{\text{high}}_{h,m} > \Delta T^{\text{low}}_{h,m}, \\ 0, & \Delta T^{\text{high}}_{h,m} \le \Delta T^{\text{low}}_{h,m}. \end{cases}$$

The numerical floor is $\varepsilon = 10^{-3}\,°\text{C}$ (used in the stochastic sampling branch to avoid zero spread).

For each spell $q = 1, \ldots, n_{s,m}$, magnitude allocation follows:

$$M_{h,m,q} = \begin{cases} \mu^\star_{h,m}, & \kappa \in \mathcal{D}_{\text{det}}, \\ X_{h,m,q}, & \kappa \notin \mathcal{D}_{\text{det}} \text{ and } b_{h,m} > a_{h,m}, \\ \mu^\star_{h,m}, & \kappa \notin \mathcal{D}_{\text{det}} \text{ and } b_{h,m} \le a_{h,m}, \end{cases}$$

$$M_{h,m,q} = \begin{cases} a_{h,m}, & M_{h,m,q} < a_{h,m}, \\ M_{h,m,q}, & a_{h,m} \le M_{h,m,q} \le b_{h,m}, \\ b_{h,m}, & M_{h,m,q} > b_{h,m}. \end{cases}$$

So each spell receives one scalar magnitude $M_{h,m,q}$; all days in that spell share that same value.

Define day-of-month anomaly allocation:

$$A_{h,m}(r) = \begin{cases} 0, & r \notin \mathcal{B}_{h,m,1} \cup \cdots \cup \mathcal{B}_{h,m,n_{s,m}}, \\ M_{h,m,q}, & r \in \mathcal{B}_{h,m,q} \text{ for the unique spell } q, \end{cases} \qquad r = 1, \ldots, L_m.$$

Because $\mathcal{B}_{h,m,q}$ are non-overlapping by construction, there is at most one active spell for each day r.

Step 18.3-F: Explicit stochastic draw used when $\kappa \notin \mathcal{D}_{\text{det}}$. For each spell q, define $X_{h,m,q}$ (the stochastic spell-anomaly draw, °C) by rejection sampling:

$$g_{\text{std}}(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right), \qquad Z_{h,m,q,t} \sim g_{\text{std}}, \quad t = 1, \ldots, 64$$

Here z is a dimensionless standard-normal coordinate (z-score), and $Z_{h,m,q,t}$ is one random draw from that $N(0,1)$ distribution. So z itself is unbounded:

$$z \in (-\infty, \infty).$$

Practical bounds are imposed only after affine transformation to anomaly space, because accepted draws must satisfy

$$a_{h,m} \leq Y_{h,m,q,t} \leq b_{h,m}.$$

Equivalently, accepted z-draws lie in

$$\frac{a_{h,m} - \mu^{\star}_{h,m}}{\sigma^{\star}_{h,m}} \leq z \leq \frac{b_{h,m} - \mu^{\star}_{h,m}}{\sigma^{\star}_{h,m}}.$$

$$Y_{h,m,q,t} = \mu^{\star}_{h,m} + \sigma^{\star}_{h,m} Z_{h,m,q,t}.$$

$$\tau_{h,m,q} = \min\left\{t \in \{1, \ldots, 64\} : \; a_{h,m} \leq Y_{h,m,q,t} \leq b_{h,m}\right\}.$$

If $\tau_{h,m,q}$ exists, set

$$X_{h,m,q} = Y_{h,m,q,\tau_{h,m,q}}.$$

If no accepted draw appears in 64 attempts, the implementation fallback is

$$X_{h,m,q} = \min\Big(\max\Big(\mu^{\star}_{h,m}, a_{h,m}\Big), \, b_{h,m}\Big).$$

So every accepted draw lies inside the physical anomaly range $[a_{h,m}, b_{h,m}]$.

Step 18.3-G: Apply heat-wave override on selected spell days. Let $r_m(d) \in \{1, \ldots, L_m\}$ be the day-of-month corresponding to day-of-year d in month m. Then

$$T^{\text{th}}_{h,m}(d) = \bar{T}_m + A_{h,m}(r_m(d)).$$

Interpretation: $T^{\text{th}}_{h,m}(d)$ is the heat-wave-threshold ambient for day d in month m. It is the day-specific lift value generated from the monthly mean $\bar{T}_m$ and allocated spell anomaly $A_{h,m}(r_m(d))$.

$$T_{\infty}(d) = \begin{cases} \max\Big(T_{\infty,\text{base}}(d), \, T^{\text{th}}_{h,m}(d), \, T_{\infty,\text{preH}}(d)\Big), & r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}, \\ T_{\infty,\text{preH}}(d), & r_m(d) \notin \mathcal{B}^{\text{all}}_{h,m}. \end{cases}$$

So 18.3 applies a single spell-day max among baseline, heat-wave threshold, and the pre-heat-wave profile. After this month-wise heat-wave override, $T_{\infty}(d)$ is passed into Section 18.4.

In implementation, random sampling uses a fixed seed (`spellAnomalyRngSeed`) for reproducibility. In this repository state, $\Delta T_{h,m}$, $\Delta T^{\text{low}}_{h,m}$, $\Delta T^{\text{high}}_{h,m}$, and $\sigma_{\Delta T,h,m}$ are NOAA-derived from 2016–2025 daily station data (`USW00014740`) with spell-level anomalies grouped by spell start month.

18.4 Daily heating and cooling loads (239, 240)

For each representative day d, after final ambient assignment $T_\infty(d)$, I rebuilt the bin thermal-loss model and computed required loads:

$$Q_{\text{req},h}(d) = UA_{\text{bin}} \max(T_{h,\text{set}} - T_\infty(d),\, 0)\,,$$

$$Q_{\text{req},c}(d) = UA_{\text{bin}} \max(T_\infty(d) - T_{c,\text{set}},\, 0)\,.$$

The first equation activates only when ambient is below heating setpoint; the second activates only when ambient is above cooling setpoint.

18.5 Capacity re-solve and duty application (241–246)

For each day, I did not reuse fixed design capacities. Instead, I re-solved the selected heating or cooling operating point at that day's ambient:

$$Q_{h,\text{cap}}(d) = Q_{\text{to bed},h}(T_\infty(d)), \quad P_{h,\text{elec}}(d) = P_h(T_\infty(d)),$$

$$Q_{c,\text{cap}}(d) = -Q_{\text{to bed},c}(T_\infty(d)), \quad P_{c,\text{elec}}(d) = P_{\text{spot cooler}}(d) + P_{\text{assist blower}}(d).$$

Identity clarification for these symbols: $Q_{\text{to bed},h}(T_\infty(d))$ and $Q_{\text{to bed},c}(T_\infty(d))$ are the signed net bed heat-transfer rates returned by the heating and cooling day-ambient reference-point re-solves, respectively. $P_h(T_\infty(d))$ is the heating electrical input from the heating re-solve. $P_{\text{spot cooler}}(d)$ and $P_{\text{assist blower}}(d)$ are the two cooling electrical components returned by the cooling re-solve. Therefore $Q_{h,\text{cap}}(d)$ and $Q_{c,\text{cap}}(d)$ are positive capacity magnitudes used for duty calculation, and $P_{h,\text{elec}}(d)$, $P_{c,\text{elec}}(d)$ are the corresponding mode electric powers used for $E_h(d)$, $E_c(d)$.

What "re-solve" means mathematically (explicit). For each day d, the model re-runs the selected heating/cooling reference physics at ambient $T_\infty(d)$, not at a fixed design ambient.

Heating electrical power $P_h(T_\infty(d))$. Wire resistance is temperature-corrected:

$$R_{\text{tube},g}(d) = \rho_{20}\Big[1 + \alpha\Big(\bar{T}_{w,g}(d) - 20\Big)\Big]\,\frac{L_{w,g}}{A_{w,g}},$$

then branch/tube current and power are

$$I_{\text{tube},g}(d) = \frac{V}{N_{s,g} R_{\text{tube},g}(d)}, \qquad P_{\text{tube},g}(d) = I^2_{\text{tube},g}(d) R_{\text{tube},g}(d),$$

so coil-only heating power is

$$P_{h,\text{coil}}(d) = \sum_g n_{\text{tube},g}\, P_{\text{tube},g}(d).$$

The heating blower uses the same flow/pressure/power closure structure as the cooling assist blower. For each blower $x \in \{\text{hb}, \text{ab}\}$, define

$$\Delta p_{\text{avail},x}(\dot{V}) = \max\left(\Delta p_{\text{shutoff},x}\left(1 - \frac{\dot{V}}{\dot{V}_{\text{free},x}}\right), 0\right),$$

with $\dot{V}_{\text{free},x}$ taken from config when provided, otherwise inferred from rated point data as

$$\dot{V}_{\text{free},x} = \begin{cases} \dot{V}_{\text{rated},x}\,\dfrac{\Delta p_{\text{shutoff},x}}{\Delta p_{\text{shutoff},x} - \Delta p_{\text{rated},x}}, & \Delta p_{\text{shutoff},x} > \Delta p_{\text{rated},x}, \\ \dot{V}_{\text{rated},x}, & \text{otherwise.} \end{cases}$$

This comes from a two-point linear fan curve through

$$(\dot{V}, \Delta p) = (0, \Delta p_{\text{shutoff},x}) \quad \text{and} \quad (\dot{V}, \Delta p) = (\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x}).$$

Its slope is

$$m_x = \frac{\Delta p_{\text{rated},x} - \Delta p_{\text{shutoff},x}}{\dot{V}_{\text{rated},x}},$$

so

$$\Delta p_x(\dot{V}) = \Delta p_{\text{shutoff},x} + m_x \dot{V}.$$

Setting $\Delta p_x(\dot{V}_{\text{free},x}) = 0$ gives Eq. 18.5C-1a2. Physically, $\Delta p_{\text{shutoff},x}$ should be larger than $\Delta p_{\text{rated},x}$; if not, this linear extrapolation is not reliable, so code falls back to $\dot{V}_{\text{free},x} = \dot{V}_{\text{rated},x}$. Interpretation in plain terms: $\Delta p_{\text{shutoff},x}$ and $\Delta p_{\text{rated},x}$ are two datasheet points on the *same* blower static-pressure-vs-flow curve, both as pressure-rise values (Pa), not absolute pressures. $\Delta p_{\text{shutoff},x}$ is the zero-flow endpoint (blocked outlet), and $\Delta p_{\text{rated},x}$ is the pressure at the rated flow endpoint; those two points define the linear slope used for $\Delta p_{\text{avail},x}(\dot{V})$ and $\dot{V}_{\text{free},x}$. Construction sequence used in implementation: if datasheet $\dot{V}_{\text{free},x}$ is provided, the line is built directly from $(0, \Delta p_{\text{shutoff},x})$ and $(\dot{V}_{\text{free},x}, 0)$; if $\dot{V}_{\text{free},x}$ is not provided, it is inferred from $(\dot{V}_{\text{rated},x}, \Delta p_{\text{rated},x})$ and $\Delta p_{\text{shutoff},x}$, then the same $(0, \Delta p_{\text{shutoff},x})$-to-$(\dot{V}_{\text{free},x}, 0)$ line is used.

$$R_x(\dot{V}) = \Delta p_{\text{sys},x}(\dot{V}) - \Delta p_{\text{avail},x}(\dot{V}),$$

$$\Delta p_{\text{sys,ab}}(\dot{V}) = \Delta p_{\text{dist},c}(\dot{V}) + p_{\text{in},c}(\dot{V}),$$

$$\Delta p_{\text{dist},c} = \Delta p_{\text{header}} + \Delta p_{\text{header}\to\text{splitter}} + \Delta p_{\text{splitter body}} + \Delta p_{\text{connector fric}} + \Delta p_{\text{connector}\to\text{branch}},$$

where $p_{\text{in},c}$ is solved by the closed-end inlet residual

$$R_{\text{inlet}}(p_{\text{in},c}) = \dot{m}_{\text{rem},N_{\text{seg}}+1}(p_{\text{in},c}) = 0.$$

At segment s, the coupled release/backpressure closure used inside that solve is

$$\dot{m}_{\text{rel},s} = \min\left(C_d A_{h,s}\sqrt{2\rho_s \max(p_s - p_{\text{bed},s}, 0)}, \dot{m}_{\text{rem},s}\right),$$

$$p_{\text{bed},s} = p_{\text{out,bnd}} + \Delta p_{\text{Ergun},s},$$

$$\Delta p_{\text{Ergun},s} = L_s\left[\frac{150(1-\varepsilon)^2}{\varepsilon^3}\frac{\mu_s u_s}{d_p^2} + \frac{1.75(1-\varepsilon)}{\varepsilon^3}\frac{\rho_s u_s^2}{d_p}\right], \qquad u_s = \frac{\dot{m}_{\text{rel},s}}{\rho_s A_{\text{trib},s}},$$

$$\dot{m}_{\text{rem},s+1} = \dot{m}_{\text{rem},s} - \dot{m}_{\text{rel},s}, \qquad p_{s+1} = \max(p_s - \Delta p_{\text{tube},s},\, p_{\text{bed},s}).$$

For heating, the blower-side required pressure is

$$\Delta p_{\text{sys,hb}}(\dot{V}) = \Delta p_{\text{dist,common}}(\dot{V}) + \Delta p_{\text{path,ctrl}}(\dot{V}),$$

$$\Delta p_{\text{path},g} = \Delta p_{\text{dist,private},g} + \Delta p_{\text{heater,tube},g} + \Delta p_{\text{heater,coil},g} + \Delta p_{\text{heater}\to\text{aer},g} + p_{\text{in,aer},g},$$

$$\Delta p_{\text{path,ctrl}} = \Delta p_{\text{path},g_{\text{ctrl}}}, \qquad \Delta p_{\text{path},g_{\text{ctrl}}} \geq \Delta p_{\text{path},g} \ \text{ for every group } g.$$

The term $p_{\text{in,aer},g}$ is solved by the same closed-end residual form as $p_{\text{in},c}$, so porous-bed backpressure enters both cooling and heating pressure balances.

$$\mathcal{F}_x(d) = \left\{\dot{V} \in [0, \dot{V}_{\text{req},x}(d)] \ : \ R_x(\dot{V}) \leq 0\right\},$$

$$\dot{V}_{\text{feas},x}(d) = \begin{cases} \max \mathcal{F}_x(d), & \mathcal{F}_x(d) \neq \varnothing, \\ 0, & \mathcal{F}_x(d) = \varnothing. \end{cases}$$

$$\dot{V}_{\text{ach},x}(d) = \begin{cases} \dot{V}_{\text{req},x}(d), & \text{if blower } x \text{ is disabled, fan data are unavailable, or } R_x(\dot{V}_{\text{req},x}(d)) \leq 0, \\ \dot{V}_{\text{feas},x}(d), & \text{otherwise.} \end{cases}$$

$$\Delta p_{\text{op},x}(d) = \begin{cases} 0, & \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) \leq 0, \\ \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)), & 0 < \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) < \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)), \\ \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)), & \Delta p_{\text{sys},x}(\dot{V}_{\text{ach},x}(d)) \geq \Delta p_{\text{avail},x}(\dot{V}_{\text{ach},x}(d)). \end{cases}$$

$$P_{\text{shaft},x}(d) = \dot{V}_{\text{ach},x}(d)\, \Delta p_{\text{op},x}(d).$$

If total efficiency is provided, electric power is

$$P_{x,\text{elec}}(d) = \min\left(P_{\text{rated},x}, P_{\text{idle},x} + \frac{P_{\text{shaft},x}(d)}{\eta_x} \right).$$

If η_x is unavailable, the fallback used in code is

$$P_{x,\text{elec}}(d) = \min\left(P_{\text{rated},x}, P_{\text{idle},x} + (P_{\text{rated},x} - P_{\text{idle},x}) \min\left(\max\left(\frac{\dot{V}_{\text{ach},x}(d)}{\dot{V}_{\text{free},x}}, 0 \right), 1 \right) \right).$$

The full implemented logic is

$$P_{x,\text{elec}}(d) = \begin{cases} 0, & \text{if blower } x \text{ is disabled,} \\ & \text{or } \dot{V}_{\text{ach},x}(d) \leq 0 \\ \min\left(P_{\text{rated},x},\ P_{\text{idle},x} + \dfrac{P_{\text{shaft},x}(d)}{\eta_x} \right), & \text{if } \eta_x \text{ is provided,} \\ \min\left(P_{\text{rated},x}, P_{\text{idle},x} + (P_{\text{rated},x} - P_{\text{idle},x}) \min\left(\max\left(\dfrac{\dot{V}_{\text{ach},x}(d)}{\dot{V}_{\text{free},x}}, 0 \right), 1 \right) \right), & \text{otherwise.} \end{cases}$$

So $P_{\text{idle},x}$ is baseline blower electrical draw while running at low flow (motor/controller/parasitic overhead before pressure-flow work), not an always-on term. In the implementation it is read from `idlePower_W`, clamped to be nonnegative, and it contributes only when blower is enabled and solved flow is positive. For calibration, use a measured low-flow electrical point and keep $0 \leq P_{\text{idle},x} \leq P_{\text{rated},x}$. Therefore heating electrical power is

$$P_h(T_\infty(d)) = P_{h,\text{coil}}(d) + P_{\text{hb,elec}}(d).$$

Signed heating $Q_{\text{to bed},h}(T_\infty(d))$. For each active branch b and axial segment s:

$$Q_{\text{wall},b,s} = \dot{m}_{\text{mean},b,s} c_{p,b,s} (T_{\text{air,in},b,s} - T_{\text{air,out},b,s}),$$

$$Q_{\text{jet},b,s} = \dot{m}_{\text{rel},b,s} c_{p,b,s} (T_{\text{rel},b,s} - T_{\text{bed,ref}}),$$

$$Q_{\text{evap},b,s} = \dot{m}_{\text{evap},b,s}\, h_{fg,b,s},$$

and the branch/segment balance is accumulated as

$$Q_{\text{to bed},h}(T_\infty(d)) = \sum_{b=1}^{N_b} \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},b,s} + Q_{\text{jet},b,s} - Q_{\text{evap},b,s} \right).$$

Spot-cooler and assist-blower powers.

$$\dot{V}_{\text{rated}} = 0.00047194745\, V_{\text{rated,CFM}}, \qquad \dot{m}_{\text{rated}}(d) = \rho_{\text{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\text{rated}},$$

$$\dot{Q}_{\text{rated,sens}}(d) = \max\Big(\dot{Q}_{\text{rated,total}} - \dot{Q}_{\text{rated,lat}}(d),\, 0\Big),$$

$$T_{\text{target,derived}}(d) = T_\infty(d) - \frac{\dot{Q}_{\text{rated,sens}}(d)}{\dot{m}_{\text{rated}}(d) c_p(d)} \quad \text{when } \dot{m}_{\text{rated}}(d) c_p(d) > 0,$$

$$T_{\text{target}}(d) = \begin{cases} T_{\text{target,derived}}(d), & \text{if } \chi_{\text{derive}} = 1 \text{ and } T_{\text{target,derived}}(d) \text{ is finite,} \\ T_{\text{target,cfg}}, & \text{if a configured target is provided,} \\ T_\infty(d), & \text{otherwise,} \end{cases}$$

This matches code behavior: enabling `deriveTargetFromRatedSpecs` makes the derived target override the configured target when derivation is finite.

$$\dot{m}(d) = \rho_{\text{air}}(T_\infty(d), p_\infty)\, \dot{V}_{\text{ach,ab}}(d),$$

$$\dot{m}_{\text{dry}}(d) = \frac{\dot{m}(d)}{1 + w_{\text{in}}(d)}.$$

Here $\dot{V}_{\text{ach,ab}}(d)$ is the achieved assist-blower flow from the residual closure $R_{\text{ab}}(\dot{V}) = \Delta p_{\text{sys,ab}}(\dot{V}) - \Delta p_{\text{avail,ab}}(\dot{V})$, solved over $[0, \dot{V}_{\text{req,ab}}(d)]$.

$$\dot{Q}_{\text{spot,req}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{target}}(d), 0),$$

$$\dot{Q}_{\text{spot,act}}(d) = \min\Big(\dot{Q}_{\text{spot,req}}(d), \dot{Q}_{\text{spot,max}}(d)\Big),$$

$$\dot{Q}_{\text{spot,max}}(d) = \begin{cases} Q_{\text{spot,max,cfg}}, & \text{if } \texttt{maxCoolingCapacity_W} \text{ is provided,} \\ \max\Big(\dot{Q}_{\text{rated,total}} - \dot{Q}_{\text{rated,lat}}(d),\, 0\Big), & \text{if rated-spec path is used,} \\ +\infty, & \text{otherwise,} \end{cases}$$

$$\dot{Q}_{\text{rated,total}} = 0.29307107\, C_{\text{rated,BTU/h}},$$
$$\dot{m}_{\text{cond,rated}} = \frac{0.473176473\, D_{\text{rated,pint/day}}}{86400},$$
$$\dot{Q}_{\text{rated,lat}}(d) = \dot{m}_{\text{cond,rated}}\, h_{fg}(T_\infty(d)).$$

$$\dot{Q}_{\text{sensible}}(d) = \dot{m}(d) c_p(d) \max(T_\infty(d) - T_{\text{supply,actual}}(d), 0) = \dot{Q}_{\text{spot,act}}(d).$$

$$P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d).$$

$$\text{COP} = \frac{\dot{Q}_{\text{cool}}}{P_{\text{in}}(d)},$$

$$\text{COP}_{\text{sensible}} = \frac{\dot{Q}_{\text{sensible}}(d)}{P_{\text{in}}(d)},$$

$$\text{COP} = \frac{\text{EER}}{3.412}.$$

$$P_{\text{spot cooler}}(d) = \frac{\dot{Q}_{\text{spot,act}}(d)}{\text{COP}},$$

In this model, the configured COP in the equation above is interpreted as sensible COP because $\dot{Q}_{\text{spot,act}}(d)$ is sensible load. and assist-blower power is the blower electrical result above:

$$P_{\text{assist blower}}(d) = P_{\text{ab,elec}}(d).$$

At this solved $\dot{V}_{\text{ach}}(d)$, the same segment energy balance gives

$$Q_{\text{to bed},c}(T_\infty(d)) = \sum_{s=1}^{N_{\text{seg}}} \left(Q_{\text{wall},s} + Q_{\text{jet},s} - Q_{\text{evap},s}\right),$$

which is usually negative in cooling mode (heat removed from bed).

Matrix solve vs residual solve (clarification). The two-node temperature step is a direct linear solve, not an iterative matrix residual solve:

$$\mathbf{A}(d)\mathbf{T}(d) = \mathbf{b}(d), \qquad \mathbf{T}(d) = \mathbf{A}(d)^{-1}\mathbf{b}(d).$$

The iterative residual solves in this layer are scalar bisections: $R_x(\dot{V})$ for blower/system matching ($x \in \{\text{hb}, \text{ab}\}$), $R_{\text{inlet}}(p_{\text{in}}) = \dot{m}_{\text{remaining}}(p_{\text{in}})$ for closed-end inlet pressure, and $R_{\text{bed ref}}(T_{\text{ref}}) = \frac{1}{2}(T_b + T_t) - T_{\text{ref}}$ for cooling branch bed-reference closure.

Variables used in the equations above (explicit definitions).

- d: representative day index $(1, \ldots, 365)$.
- g: heater-tube group index; $n_{\text{tube},g}$ is the number of identical tubes in group g.
- b: aeration branch index $(1, \ldots, N_b)$.
- s: axial segment index $(1, \ldots, N_{\text{seg}})$.
- $R_{\text{tube},g}(d)$: electrical resistance of one tube in group g at day-d operating temperature.
- ρ_{20}: wire resistivity at $20\,^\circ\text{C}$.
- α: wire temperature coefficient of resistivity.
- $\bar{T}_{w,g}(d)$: representative wire temperature for group g, day d.
- $L_{w,g}$, $A_{w,g}$: wire length and wire cross-sectional area for group g.
- V: applied electrical voltage across each series string.
- $N_{s,g}$: number of tube elements in series in group g.
- $I_{\text{tube},g}(d)$: current through one tube/string in group g.
- $P_{\text{tube},g}(d)$: electrical power of one tube/string in group g.
- $P_{h,\text{coil}}(d)$: total heater-coil electrical power from all tube groups at day d.
- $P_{\text{hb,elec}}(d)$: heating-blower electrical power at day d.
- $P_h(T_\infty(d))$: total heating electrical input at ambient $T_\infty(d)$, equal to $P_{h,\text{coil}}(d) + P_{\text{hb,elec}}(d)$.
- $Q_{\text{wall},b,s}$: sensible heat transfer from branch airflow to bed-side walls in branch b, segment s.

- $\dot{m}_{\text{mean},b,s}$: mean airflow through segment s of branch b.
- $c_{p,b,s}$: specific heat of segment airflow in branch b, segment s.
- $T_{\text{air,in},b,s}$, $T_{\text{air,out},b,s}$: segment inlet/outlet air temperatures.
- $Q_{\text{jet},b,s}$: sensible heat carried by released jet flow relative to bed reference temperature.
- $\dot{m}_{\text{rel},b,s}$: mass flow released through perforations into bed in branch b, segment s.
- $T_{\text{rel},b,s}$: release/jet air temperature to the bed in branch b, segment s.
- $T_{\text{bed,ref}}$: branch-level bed reference temperature used for local segment balances.
- $Q_{\text{evap},b,s}$: latent heat sink due to evaporation in branch b, segment s.
- $\dot{m}_{\text{evap},b,s}$: evaporation mass flow rate in branch b, segment s.
- $h_{fg,b,s}$: latent heat of vaporization used in branch b, segment s.
- $Q_{\text{to bed},h}(T_\infty(d))$: net signed heating-mode bed heat-transfer rate; positive means heat delivered to bed.
- $\dot{Q}_{\text{spot,req}}(d)$: requested sensible spot-cooling load at day d.
- $\dot{m}(d)$, $c_p(d)$: day-d cooling-branch mass flow and effective specific heat.
- $T_{\text{target,cfg}}$: configured spot-cooler target from `targetSupplyAir_C`.
- χ_{derive}: derivation toggle from `deriveTargetFromRatedSpecs` (1 on, 0 off).
- $V_{\text{rated,CFM}}$: spot-cooler rated airflow from configuration.
- $\dot{V}_{\text{rated}}$: rated airflow converted to m^3/s.
- p_∞: day ambient absolute pressure used in air-property evaluation.
- $\rho_{\text{air}}(T, p)$: moist-air density function used to map rated airflow to rated mass flow.
- $\dot{V}_{\text{ach,ab}}(d)$: achieved cooling assist-blower flow from the blower/network residual solve.
- $\dot{m}_{\text{rated}}(d)$: rated mass flow at day-d ambient properties.
- $\dot{m}(d)$: total moist-air mass flow through the spot-cooler branch, computed as $\rho_{\text{air}}(T_\infty(d), p_\infty)\dot{V}_{\text{ach,ab}}(d)$.
- $w_{\text{in}}(d)$: inlet humidity ratio used to convert moist-air to dry-air basis.
- $\dot{m}_{\text{dry}}(d)$: dry-air mass flow, $\dot{m}(d)/(1 + w_{\text{in}}(d))$.
- $\dot{Q}_{\text{rated,sens}}(d)$: rated sensible capacity used by the target-derivation path.
- $T_{\text{target,derived}}(d)$: target supply temperature derived from rated sensible capacity and rated mass flow.
- $T_{\text{target}}(d)$: effective day-d spot-cooler target used in $\dot{Q}_{\text{spot,req}}(d)$.
- $\dot{Q}_{\text{spot,max}}(d)$: available spot-cooler sensible capacity at day d.

- $\dot{Q}_{\text{spot,act}}(d)$: actual spot-cooler sensible load after capacity cap.
- $T_{\text{supply,actual}}(d)$: actual spot-cooler supply-air temperature after capacity capping.
- $\dot{Q}_{\text{sensible}}(d)$: sensible (dry-bulb) cooling rate, defined as $\dot{m}(d)c_p(d)\max(T_\infty(d)-T_{\text{supply,actual}}(d),0)$, equal to $\dot{Q}_{\text{spot,act}}(d)$ in this model.
- $P_{\text{in}}(d)$: spot-cooler electrical input power used in COP definitions; in this model $P_{\text{in}}(d) \equiv P_{\text{spot cooler}}(d)$.
- COP: coefficient of performance used to convert sensible load to electric power.
- $P_{\text{spot cooler}}(d)$: spot-cooler electrical power.
- $x \in \{\text{hb}, \text{ab}\}$: blower index (hb: heating blower, ab: cooling assist blower).
- $\dot{V}_{\text{req},x}(d)$: requested blower volumetric flow for day d, blower x.
- $\mathcal{F}_x(d)$: feasible-flow set for blower x on day d, containing trial flows in $[0, \dot{V}_{\text{req},x}(d)]$ with $R_x(\dot{V}) \leq 0$.
- $\dot{V}_{\text{feas},x}(d)$: helper feasible-flow variable, equal to $\max \mathcal{F}_x(d)$ when feasible set is nonempty, else 0.
- $\dot{V}_{\text{ach},x}(d)$: achieved flow for blower x, equal to requested flow under disabled/valid-request-feasible conditions, otherwise $\dot{V}_{\text{feas},x}(d)$.
- $\Delta p_{\text{shutoff},x}$: blower x shutoff pressure, i.e., the maximum static pressure *rise* at zero flow from the blower datasheet; in code this comes from `shutoffPressure_inH2O` and is converted to Pa.
- $\dot{V}_{\text{free},x}$: blower x free-air flow (fan-curve intercept at zero pressure).
- $\dot{V}_{\text{rated},x}$: blower x rated volumetric flow from the manufacturer-rated operating point; in code this comes from `ratedFlow_CFM` and is converted to SI units.
- $\Delta p_{\text{rated},x}$: static pressure *rise* reported at the same rated operating point as $\dot{V}_{\text{rated},x}$; in code this comes from `ratedPressure_inH2O` and is converted to Pa.
- $\dot{V}_{\text{rated},x}$, $\Delta p_{\text{rated},x}$, and $\Delta p_{\text{shutoff},x}$ together define the fallback free-air endpoint inference in Eq. above when $\dot{V}_{\text{free},x}$ is not provided.
- $\Delta p_{\text{avail},x}(\dot{V})$: available blower pressure at trial flow $\dot{V}$ from the linear fan-curve relation.
- $\Delta p_{\text{sys},x}(\dot{V})$: system-required pressure drop at trial flow $\dot{V}$ for blower path x. In cooling, $\Delta p_{\text{sys,ab}} = \Delta p_{\text{dist},c} + p_{\text{in},c}$. In heating, $\Delta p_{\text{sys,hb}} = \Delta p_{\text{dist,common}} + \Delta p_{\text{path,ctrl}}$.
- $\Delta p_{\text{dist},c}$: cooling distribution loss sum $\Delta p_{\text{header}} + \Delta p_{\text{header}\to\text{splitter}} + \Delta p_{\text{splitter body}} + \Delta p_{\text{connector fric}} + \Delta p_{\text{connector}\to\text{branch}}$.
- $p_{\text{in},c}$: cooling aeration-branch inlet gauge pressure from closed-end inlet solve.
- $R_{\text{inlet}}(p_{\text{in}})$: closed-end inlet mass-balance residual $\dot{m}_{\text{rem},N_{\text{seg}}+1}(p_{\text{in}})$.
- $p_{\text{out,bnd}}$: bed outlet-boundary gauge pressure (open-top/vent/offset closure).
- $\Delta p_{\text{Ergun},s}$: porous-bed backpressure increment over path length L_s in segment s.

- ε, d_p, L_s, $A_{\mathrm{trib},s}$: bed porosity, representative particle diameter, escape-path length, and tributary porous area in segment s.
- u_s: superficial porous velocity used in Ergun pressure-drop term for segment s.
- $\Delta p_{\mathrm{tube},s}$: in-tube Darcy friction drop across segment s.
- $\Delta p_{\mathrm{dist,common}}$: heating common distribution losses (header and header-to-splitter path).
- $\Delta p_{\mathrm{dist,private},g}$: heating group-$g$ private distribution losses (splitter body, connector friction, connector-to-branch area change).
- $\Delta p_{\mathrm{heater,tube},g}$, $\Delta p_{\mathrm{heater,coil},g}$, $\Delta p_{\mathrm{heater}\to\mathrm{aer},g}$: heater-tube friction, heater-coil obstruction, and heater-to-aeration area-change losses for group g.
- $p_{\mathrm{in,aer},g}$: group-g aeration inlet gauge pressure from the same closed-end release/backpressure closure.
- $\Delta p_{\mathrm{path},g}$: full group-$g$ private path pressure requirement in heating.
- g_{ctrl}: controlling heating group (the group with the largest $\Delta p_{\mathrm{path},g}$).
- $R_x(\dot{V})$: blower/system residual for blower x, $R_x = \Delta p_{\mathrm{sys},x} - \Delta p_{\mathrm{avail},x}$.
- $\Delta p_{\mathrm{op},x}(d)$: operating pressure used for power at solved day-d flow, bounded by system-required and available pressures.
- $P_{\mathrm{shaft},x}(d)$: shaft (fluid) power for blower x, $P_{\mathrm{shaft},x} = \dot{V}_{\mathrm{ach},x}\Delta p_{\mathrm{op},x}$.
- $P_{\mathrm{idle},x}$: baseline low-flow electrical draw for blower x (motor/controller/parasitics), from `idlePower_W`; in implemented logic it is nonnegative and only applied when blower is enabled and $\dot{V}_{\mathrm{ach},x} > 0$, otherwise blower electrical power is set to 0.
- $P_{\mathrm{rated},x}$: motor rated electrical power cap for blower x.
- η_x: total blower efficiency used in P_{shaft}/η conversion for blower x.
- $P_{x,\mathrm{elec}}(d)$: blower-x electrical power at day d, computed by efficiency model (or flow-fraction fallback), then capped by $P_{\mathrm{rated},x}$.
- $P_{\mathrm{assist\,blower}}(d)$: cooling assist-blower electrical power, equal to $P_{\mathrm{ab,elec}}(d)$.
- $Q_{\mathrm{to\,bed},c}(T_\infty(d))$: net signed cooling-mode bed heat-transfer rate; typically negative (heat removed from bed).
- $\mathbf{A}(d)$, $\mathbf{T}(d)$, $\mathbf{b}(d)$: day-d two-node linear-system matrix, unknown temperature vector, and right-hand-side vector.
- $R_{\mathrm{inlet}}(p_{\mathrm{in}})$: inlet-pressure scalar residual in the closed-end pressure closure.
- $\dot{m}_{\mathrm{remaining}}(p_{\mathrm{in}})$: remaining mass-flow mismatch used in R_{inlet}.
- $R_{\mathrm{bed\,ref}}(T_{\mathrm{ref}})$: bed-reference closure residual in cooling branch.
- T_b, T_t: bottom and top bed-node temperatures used by the closure.

- T_{ref}: trial bed reference temperature in the cooling closure.

Derivation notes (why these forms are used).

- $R_{\text{tube},g}$, $I_{\text{tube},g}$, and $P_{\text{tube},g}$ follow temperature-corrected resistivity plus Ohm's law and Joule heating.
- $Q_{\text{to bed},h}$ and $Q_{\text{to bed},c}$ are first-law balances over all active segments: sensible-to-wall + sensible-by-jet − latent evaporation.
- $\dot{Q}_{\text{spot,req}}$ is a sensible-load relation $\dot{m}c_p\Delta T$; $\dot{Q}_{\text{spot,act}}$ applies equipment capacity capping.
- $P_{\text{spot cooler}} = \dot{Q}_{\text{spot,act}}/\text{COP}$ is the standard load-to-electric relation under the model COP assumption.
- $\Delta p_{\text{avail},x}(\dot{V})$ is the linear fan-curve model between shutoff and free-delivery points.
- $\Delta p_{\text{sys},x}(\dot{V})$ explicitly includes friction/minor losses and porous-bed backpressure through the inlet-pressure closure; it is not a friction-only proxy.
- $R_x(\dot{V}) = 0$ enforces operating-point consistency for each blower: system pressure requirement equals blower available pressure.
- $P_{\text{shaft},x} = \dot{V}_{\text{ach},x}\Delta p_{\text{op},x}$ converts solved operating point to fluid power; dividing by η_x and adding idle power gives electrical input, then rating cap is enforced.
- Heating electrical power is coil plus heating-blower power: $P_h = P_{h,\text{coil}} + P_{\text{hb,elec}}$.
- $\mathbf{AT} = \mathbf{b}$ is solved directly for the two thermal nodes; only the closure constraints use scalar residual bisection.

I then computed required duty:

$$D_h^*(d) = \frac{Q_{\text{req},h}(d)}{Q_{h,\text{cap}}(d)}, \qquad D_c^*(d) = \frac{Q_{\text{req},c}(d)}{Q_{c,\text{cap}}(d)},$$

and clipped applied duty to physical bounds:

$$D_h(d) = \min(\max(D_h^*(d), 0), 1), \qquad D_c(d) = \min(\max(D_c^*(d), 0), 1).$$

The generic clipped-duty form f_{day} in Eq. (244) maps to these mode-specific quantities through the active heating/cooling branch.

The applied bed load is then:

$$Q_{\text{bed}}(d) = \begin{cases} D_h(d)\, Q_{h,\text{cap}}(d), & \text{heating day,} \\ -D_c(d)\, Q_{c,\text{cap}}(d), & \text{cooling day,} \\ 0, & \text{idle day.} \end{cases}$$

If required duty exceeds unity, I preserve $D = 1$ as the applied maximum and record unmet load separately; this is how capacity shortfall is represented without violating equipment limits.

18.6 Daily bed temperatures, electricity, cost, and water (247–251)

Given applied $Q_{\text{bed}}(d)$, I solved the same two-node steady bed model used elsewhere:

$$\begin{bmatrix} UA_{b,\infty} + UA_{bt} & -UA_{bt} \\ -UA_{bt} & UA_{t,\infty} + UA_{bt} \end{bmatrix} \begin{bmatrix} T_b(d) \\ T_t(d) \end{bmatrix} = \begin{bmatrix} f_b Q_{\text{bed}}(d) + UA_{b,\infty} T_\infty(d) \\ (1 - f_b) Q_{\text{bed}}(d) + UA_{t,\infty} T_\infty(d) \end{bmatrix}.$$

I computed daily electrical energy by mode:

$$E_h(d) = \frac{P_{h,\text{elec}}(d)\, D_h(d)\, 24}{1000}, \qquad E_c(d) = \frac{P_{c,\text{elec}}(d)\, D_c(d)\, 24}{1000},$$

then

$$E_{\text{day}} = E_h(d) + E_c(d).$$

Daily cost is:

$$C(d) = c_{\text{elec}} E_{\text{day}},$$

and daily water replacement estimate is:

$$m_{\text{water}}(d) = \dot{m}_{\text{water},24h}\, D(d),$$

where $D(d)$ is the active heating or cooling duty on that day.

Section 18 feed-forward map (explicit variable dependency order)

18.1 Climate reference values and regression baseline (229, 230)**. Inputs:**

- month lengths $\{L_m\}$,
- month-midpoint reference days $\{d_i\}$ (equivalently $\{d_m\}$),
- monthly means $\{\bar{T}_m\}$,
- monthly standard deviations $\{\sigma_m\}$,
- kernel bandwidth h.

Feed-forward chain:

$$\{\bar{T}_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to T_{\infty,\text{base}}(d),$$

$$\{\sigma_m, d_i, h\} \to \delta_i(d),\ w_i(d) \to \sigma(d).$$

Outputs passed forward:

- baseline ambient profile $T_{\infty,\text{base}}(d)$,
- daily spread profile $\sigma(d)$.

18.2 Freeze-day allocation and anomaly override (231–234)**. Inputs:**

- expected freeze days F_m,
- mean freeze day $\mu_{f,m}$,
- freeze monthly center $M_{f,m}$,
- freeze anomaly bounds $\Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}$,
- freeze anomaly spread $\sigma_{\Delta T,f,m}$,
- baseline ambient $T_{\infty,\text{base}}(d)$ from 18.1.

Feed-forward chain:

$$F_m \to n_{f,m} \text{ from } \beta_{f,m} = \lfloor F_m \rfloor,\ \rho_{f,m} = F_m - \beta_{f,m},\ U^{\text{cnt}}_{f,m} \sim \text{Uniform}(0,1) \to \mathcal{S}_{f,m},$$

$\{T_{\infty,\text{base}}(d), \bar{T}_m, M_{f,m}, \Delta T^{\text{low}}_{f,m}, \Delta T^{\text{high}}_{f,m}, \sigma_{\Delta T,f,m}, \mathcal{S}_{f,m}\} \to M_{f,m},\ A_{f,m}(r) \to T_{\infty,\text{preH}}(d)$ via the 18.2-G conditional o

Output passed forward:

- pre-heat-wave ambient profile $T_{\infty,\text{preH}}(d)$.

18.3 Heat-wave allocation and anomaly override (235–238)**. Inputs:**

- expected spell starts S_m,
- expected heat-wave days H_m,
- mean start day $\mu_{h,m}$,
- heat-wave mean anomaly $\Delta T_{h,m}$,
- heat-wave anomaly bounds $\Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}$,
- heat-wave anomaly spread $\sigma_{\Delta T,h,m}$,
- pre-heat-wave profile from 18.2.

Feed-forward chain:

$$\{S_m, H_m\} \to \{n_{s,m}, n_{h,m}\} \to \mathcal{S}_{h,m} \text{ (using the same recursive set logic as 18.2)},$$

$$\{\Delta T_{h,m}, \Delta T^{\text{low}}_{h,m}, \Delta T^{\text{high}}_{h,m}, \sigma_{\Delta T,h,m}, \mathcal{S}_{h,m}\} \to \{\Delta T_{h,m,q}\}_{q=1}^{n_{s,m}},$$

$$\{T_{\infty,\text{base}}(d), \bar{T}_m, \Delta T_{h,m,q}, T_{\infty,\text{preH}}(d)\} \to T_\infty(d)$$

$$= \max\Big(T_{\infty,\text{base}}(d),\ \bar{T}_m + \Delta T_{h,m,q(d)},\ T_{\infty,\text{preH}}(d)\Big) \text{ on spell days } r_m(d) \in \mathcal{B}^{\text{all}}_{h,m}.$$

Output passed forward:

- final ambient profile $T_\infty(d)$ used in day-level thermal solves.

18.4 Daily heating and cooling loads (239, 240). Inputs:

- final ambient $T_\infty(d)$ from 18.3,
- setpoints $T_{h,\text{set}}$, $T_{c,\text{set}}$,
- day-resolved effective bin conductance terms (UA_{bin}).

Feed-forward chain:

$$\{T_\infty(d), T_{h,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},h}(d),$$
$$\{T_\infty(d), T_{c,\text{set}}, UA_{\text{bin}}\} \to Q_{\text{req},c}(d).$$

Outputs passed forward:

- required day loads $Q_{\text{req},h}(d)$, $Q_{\text{req},c}(d)$.

18.5 Capacity re-solve and duty application (241–246). Inputs:

- required day loads from 18.4,
- final ambient $T_\infty(d)$ from 18.3,
- selected heating and cooling reference points.

Feed-forward chain:

$$\{T_\infty(d), \text{heating reference}\} \to \{Q_{h,\text{cap}}(d), P_{h,\text{elec}}(d)\},$$
$$\{T_\infty(d), \text{cooling reference}\} \to \{Q_{c,\text{cap}}(d), P_{c,\text{elec}}(d)\},$$
$$\{Q_{\text{req},h}(d), Q_{h,\text{cap}}(d)\} \to D_h^*(d) \to D_h(d),$$
$$\{Q_{\text{req},c}(d), Q_{c,\text{cap}}(d)\} \to D_c^*(d) \to D_c(d),$$
$$\{D_h(d), D_c(d), Q_{h,\text{cap}}(d), Q_{c,\text{cap}}(d)\} \to Q_{\text{bed}}(d).$$

Outputs passed forward:

- applied bed load $Q_{\text{bed}}(d)$,
- mode duty $D(d)$,
- mode electrical power ($P_{h,\text{elec}}(d)$ on heating days or $P_{c,\text{elec}}(d)$ on cooling days).

18.6 Daily temperatures, electricity, cost, and water (247–251). Inputs:

- $Q_{\text{bed}}(d)$, duty, and power from 18.5,
- ambient $T_\infty(d)$ from 18.3,
- two-node conductance parameters and split fraction f_b,
- electricity price c_{elec},
- nominal daily water rate $\dot{m}_{\text{water},24h}$.

Feed-forward chain:

$$\{Q_{\text{bed}}(d), T_\infty(d), UA_{b,\infty}, UA_{t,\infty}, UA_{bt}, f_b\} \to \{T_b(d), T_t(d)\},$$
$$\{P_{h,\text{elec}}(d), D_h(d)\} \to E_h(d), \quad \{P_{c,\text{elec}}(d), D_c(d)\} \to E_c(d) \to E_{\text{day}},$$
$$\{E_{\text{day}}, c_{\text{elec}}\} \to C(d),$$
$$\{\dot{m}_{\text{water},24h}, D(d)\} \to m_{\text{water}}(d).$$

Annual aggregation closure:

$$\sum_{d=1}^{365} \{E_h(d), E_c(d), C(d), m_{\text{water}}(d)\} \to \{\text{annual energy, annual cost, annual water}\}.$$

18.7 Literature-backed microbial heat source q_{bio} for winter heating

To model partial winter reliance on in-bed microbial activity, I added an explicit volumetric source term q_{bio} in the bottom-mat reduced heating solver. The source is parameterized from oxygen uptake rate (OUR) and oxygen-to-heat conversion enthalpy, then converted to volumetric units for direct insertion into the bed energy equation.

Primary source references used in this implementation.

- Sonowal et al. report dewatered-sludge vermicomposting OUR values of 18.9 ± 0.5 mg O_2/g VS/day initially, and late-stage values as low as 2.0–3.1 mg O_2/g VS/day [R20].
- de Guardia et al. use the biodegradation heat model $H_{\text{BIO}} = r_{O_2} h_{\text{BIO}}$, and report literature h_{BIO} spans roughly 304–448 kJ/mol O_2, with 440 kJ/mol O_2 used in their calibrated runs [R19].
- Harper et al. report an empirical oxygen-to-heat slope of about 9.8 MJ/kg O_2, equivalent to 313.6 kJ/mol O_2, providing a lower empirical conversion reference [R18].
- Gea et al. show OUR is the most reliable process activity indicator under dynamic aeration monitoring, and provide broad OUR ranges across wastes that support dynamic activity modeling rather than static maturity-only values [R21].

Step 1: OUR time profile at the selected winter operating stage. I represent process-stage decay from an initial value to a mature value with:

$$\text{OUR}(t_d) = \text{OUR}_{\text{mature}} + (\text{OUR}_{\text{initial}} - \text{OUR}_{\text{mature}}) \exp(-k_d t_d),$$

where t_d is operating day in days and k_d is in day^{-1}. The default $k_d = 0.0353$ day^{-1} is the mean implied by reported 45-day fold reductions from Sonowal et al. [R20].

Step 2: Convert OUR to heat per unit VS mass. Using the selected scenario enthalpy h_{O_2} (kJ/mol O_2):

$$q_{\text{bio,VS}}\ [\text{W/kgVS}] = \text{OUR}\ [\text{mg } O_2/\text{gVS/day}] \cdot \frac{h_{O_2}}{32} \cdot \frac{1000}{86400}.$$

The script exposes scenario bounds:

- conservative: $h_{O_2} = 313.6$ kJ/mol O_2 (Harper reference),
- base: $h_{O_2} = 376.8$ kJ/mol O_2 (midpoint working value),
- aggressive: $h_{O_2} = 440$ kJ/mol O_2 (de Guardia working value).

Step 3: Convert to volumetric source term for the bed field solve.

$$q_{\text{bio,vol}}\ [\text{W/m}^3] = q_{\text{bio,VS}} \cdot \rho_{\text{bulk}} \cdot f_{\text{VS,wet}},$$

where ρ_{bulk} is wet bulk density and $f_{\text{VS,wet}}$ is wet-mass VS fraction.

Step 4: Apply local temperature response and clipping. I use a bounded Q_{10} scaling:

$$f_T(T) = \text{clip}\Big(Q_{10}^{(T-T_{\text{ref}})/10}, f_{T,\text{min}}, f_{T,\text{max}}\Big), \qquad q_{\text{bio,local}} = q_{\text{bio,vol}}\, f_T(T).$$

Defaults are $Q_{10} = 2.0$, $T_{\text{ref}} = 22°\text{C}$, and clip 0.35–1.65, preventing runaway positive feedback in the explicit relaxation loop.

Step 5: Insert into the bed PDE update. For the 2D reduced field, the interior update adds:

$$\frac{\partial T}{\partial t} \supset \frac{q_{\text{bio,local}}}{\rho c_p},$$

so the implemented source competes directly with evaporation and wall-loss sinks in the same units.

Step 6: Net winter external-heating requirement. For duty and feasibility calculations, the winter external requirement is decomposed as:

$$Q_{\text{loss,gross}} = UA\,(T_{\text{set}} - T_\infty), \qquad Q_{\text{ext,req}} = \max(Q_{\text{loss,gross}} - Q_{\text{bio,ref}},\, 0)\,,$$

where $Q_{\text{bio,ref}}$ is the temperature-adjusted reference bioheat at T_{set}. This is now printed explicitly in the heating summary as gross loss, microbial credit, and residual external requirement.

18.8 Practical interpretation for report writing

The implemented q_{bio} model is intentionally transparent and auditable:

- every conversion step is explicit (OUR → W/kgVS → W/m^3),
- all literature-sensitive terms are configuration parameters (OUR bounds, h_{O_2} scenario, decay rate, VS fraction),
- solver outputs now separate mat heat and microbial heat contributions,
- the net-load decomposition is reported explicitly, so winter feasibility claims can be traced to assumptions instead of hidden fitting.

Implementation trace (how I actually executed it in code)

I executed this workflow through the following function chain:

- configuration and climate defaults: `year_round_analysis_config(...)`,
- integer allocation and event placement: month-wise stochastic-rounding equations from 18.2/18.3, plus `gaussian_weighted_day_selection(...)` and `heatwave_day_mask(...)`,
- spell anomaly sampling and overlays: `sample_triangular_mean_matched(...)` (freeze default), `sample_truncated_normal(...)` (heat-wave and legacy freeze mode), `circular_gaussian_kernel_reg` `build_representative_climate_year(...)`,

- microbial source assembly for winter heating: `_bioheat_base_terms(...)`, `_bioheat_temperature_factor(...`
  `_bioheat_reference_power_w(...)`,
- daily simulation task: `simulate_year_round_day_task(...)`,
- ambient-specific re-solve of operating points: `simulate_heating_reference_point(...)` and `simulate_summer_cooling_point(...)`,
- bottom-mat field solve with explicit q_{bio} insertion: `simulate_bottom_mats_state(...)`,
- daily and annual aggregation: `build_year_round_payload(...)`,
- report export: `write_year_round_summary(...)`.

Reproducibility notes

To ensure reproducibility, I preserved deterministic timing-selection equations (fixed ranking rules over remaining start-day candidates), then used seeded stochastic spell sampling (`spellAnomalyRngSeed`) with explicit bound-enforcement equations, plus explicit annual summary totals for freeze days, heat-wave starts, heat-wave days, energy, cost, water, and unmet-load equivalent.

In summary, the Section 18 implementation is an equation-driven annual operations layer with deterministic timing and seeded stochastic spell intensities: smooth baseline climate from monthly reference values, event overlays with bounded spell anomalies, ambient-specific capacity re-solving, physically bounded duty application, and day-level thermo-economic output accumulation.